Tobit Models

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Outline

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Introduction

Limited dependent variable

- There are two types of data censoring. One is called "data censoring", such as top coding of income. The other is called "corner solution outcomes", where the outcome variable piles up at one or two corners, such as women's annual labor supply which piles up at zero hour.
- "Do I always observe the outcome variable?" For corner solution, the answer is "yes". For data censoring, the answer is "no".
- We focus on corner solution case here.

- Linear model can still be applied. However there are a few drawbacks.
  1. Fitted values can be negative;
  2. The "real" partial effects are unlikely to be constant throughout the range of x as implied by linear model;

Type I Tobit model

The model

- Consider the case where $y \geq 0$ has a corner at zero. A latent variable formulation helps, where y^* is the latent variable:

$$y^* = x'\beta + u, \quad y = \max(0, y^*).$$

  y^* is notation device. You can understand y^* as the "desired" amount of the outcome.

- Type I Tobit assumes normality of the latent error u:

$$u|x \sim \mathcal{N}(0, \sigma^2).$$

Quantities of interest

- Let $w = 1(y > 0)$. Then

$$\Pr(w = 1 \mid x) = \Phi(x'\beta/\sigma).$$

- This is a probit model with parameters β/σ. The partial effects is

$$\frac{\partial \Pr(y > 0 \mid x)}{\partial x_j} = (\beta_j/\sigma)\phi(x'\beta/\sigma).$$

- The following result is very useful not only here but also for sample selection models.

  Lemma 1 *If* $z \sim \mathcal{N}(0,1)$ *then*

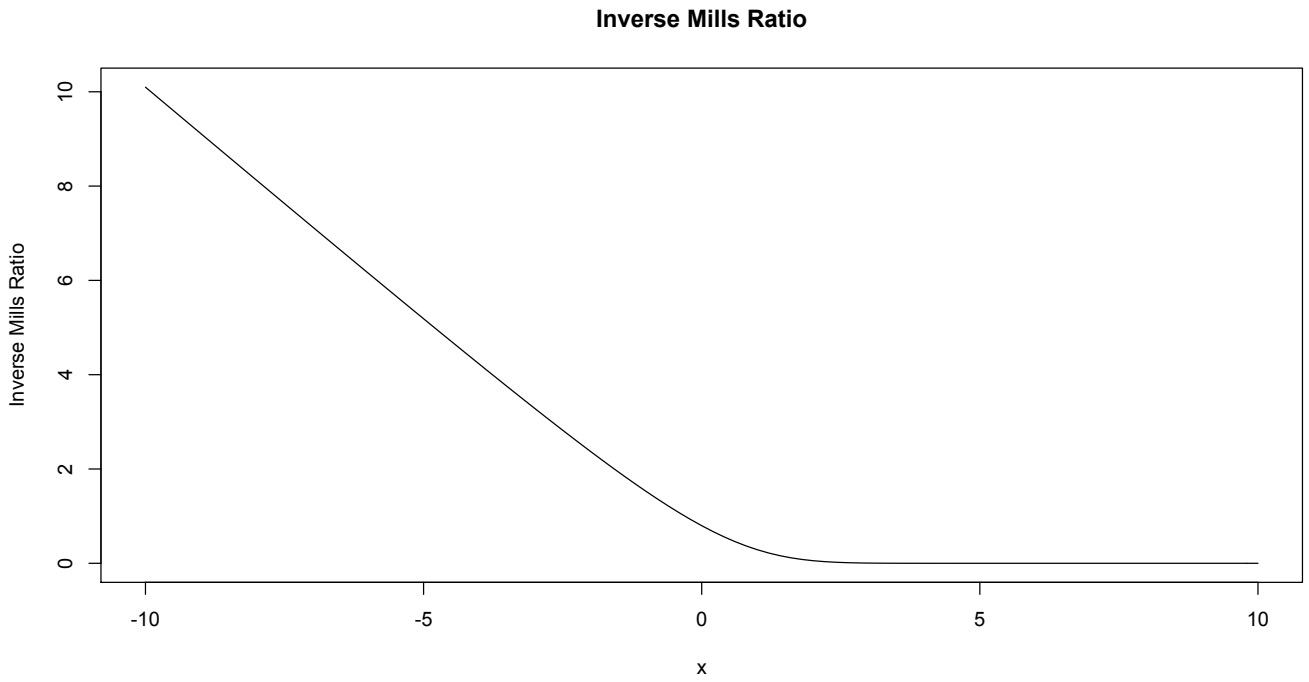
$$\mathrm{E}(z \mid z > c) = \phi(c)/(1 - \Phi(c)) \text{ or } \mathrm{E}(z \mid z > -c) = \phi(c)/\Phi(c).$$

- Because $y = \max(0, x'\beta + u)$, $y > 0$ iff $x'\beta + u > 0$. We have

$$\begin{aligned}
\mathrm{E}(y \mid x, y > 0) &= x'\beta + \mathrm{E}(u \mid x'\beta + u > 0) \\
&= x'\beta + \sigma\,\mathrm{E}(u/\sigma \mid u/\sigma > -x'\beta/\sigma) \\
&= x'\beta + \sigma\left(\frac{\phi(-x'\beta/\sigma)}{1 - \Phi(-x'\beta/\sigma)}\right) \\
&= x'\beta + \sigma\left(\frac{\phi(x'\beta/\sigma)}{\Phi(x'\beta/\sigma)}\right) \\
&\equiv x'\beta + \sigma\lambda(x'\beta/\sigma).
\end{aligned}$$

  The function $\lambda(z) = \phi(z)/\Phi(z)$ is called the *inverse Mills ratio*.

- $\lambda(z)$ is *strictly decreasing*, and looks *linear* over much of its range.
- Almost linear feature turns out to be important latter.
- $\lim_{z\to\infty} \lambda(z) = 0$ and $\lim_{z\to-\infty} \lambda(z) = \infty$.
- $\partial\lambda(z)/\partial z = -\lambda(z)(z + \lambda(z))$.



- Letting $w = 1(y > 0)$ be a dummy variable, it follows from the law of iterated expectation that

$$
\begin{aligned}
\mathrm{E}(y \mid x) &= \mathrm{E}(\mathrm{E}(y \mid x, w) \mid x) \\
&= \Pr(w = 0 \mid x) \cdot 0 + \Pr(w = 1 \mid x) \cdot \mathrm{E}(y \mid x, y > 0) \\
&= \Phi(x'\beta/\sigma) \cdot (x'\beta + \sigma\lambda(x'\beta/\sigma)) \\
&= \Phi(x'\beta/\sigma)x'\beta + \sigma\phi(x'\beta/\sigma).
\end{aligned}
$$

Partial effects

- Partial effects on $\Pr(y > 0 \mid x)$:

$$\frac{\partial \Pr(y > 0 \mid x)}{\partial x_j} = (\beta_j/\sigma)\phi(x'\beta/\sigma).$$

- Partial effects on $\mathrm{E}(y \mid x, y > 0)$:

$$\begin{aligned}\frac{\partial \mathrm{E}(y \mid x, y > 0)}{\partial x_j} &= \beta_j - \beta_j \lambda(x'\beta/\sigma)(x'\beta/\sigma + \lambda(x'\beta/\sigma)) \\ &= \beta_j \left(1 - \lambda(x'\beta/\sigma)\left[x'\beta/\sigma + \lambda(x'\beta/\sigma)\right]\right) \\ &\equiv \beta_j \theta(x'\beta/\sigma).\end{aligned}$$

- The ratio of the partial effects of x_j and x_k (assuming both are continuous) is β_j/β_k.
- β_j cannot be used directly as partial effect. Indeed, this is good because we allow the partial effect to vary across x.

- It can be shown that $\theta(c) \equiv 1 - \lambda(c)[c + \lambda(c)] = \mathrm{Var}(z \mid z > -c)$, where $z \sim \mathcal{N}(0,1)$. Also, it can be shown that $\mathrm{Var}(z \mid z > -c) < 1$ for any c. So $\theta(c) \in (0,1)$. Hence,

$$\left| \frac{\partial \mathrm{E}(y \mid x, y > 0)}{\partial x_j} \right| < |\beta_j|.$$

- β_j gives the direction of partial effect, but must be scaled *down*.

- As for the partial effect on $\mathrm{E}(y \mid x)$, we of course can take the derivative of its closed form.
- We can also use the chain rule,

$$\mathrm{E}(y \mid x) = \Pr(y > 0 \mid x) \cdot \mathrm{E}(y \mid x, y > 0).$$

We have

$$\frac{\partial \mathrm{E}(y \mid x)}{\partial x_j} = \frac{\partial \Pr(y > 0 \mid x)}{\partial x_j} \cdot \mathrm{E}(y|x, y > 0) + \Pr(y > 0|x) \cdot \frac{\partial \mathrm{E}(y \mid x, y > 0)}{\partial x_j}.$$

The conclusion is

$$\frac{\partial \mathrm{E}(y \mid x)}{\partial x_j} = \Phi(x'\beta/\sigma)\beta_j = \Pr(y > 0 \mid x)\beta_j.$$

- This partial effect has two folds.
- β_j needs to be scaled by $\Pr(y > 0 \mid x)$ (in labor force participation example, this is the likelihood of going to labor market!).

Estimation

- We have shown that both $\mathrm{E}(y \mid x, y > 0)$ and $\mathrm{E}(y \mid x)$ does not equal $x'\beta$.
- OLS of y_i on x_i using either the full sample or the sample with $y_i > 0$ does not consistently estimate β.
- MLE is standard. The density of y given x is

$$f(y|x) = \left(1 - \Phi(x'\beta/\sigma)\right)^{1(y=0)} \left(\sigma^{-1}\phi((y - x'\beta)/\sigma)\right)^{1(y>0)}.$$

- The log likelihood is

$$\ell_i(\beta, \sigma) = 1(y_i = 0)\log(1-\Phi(x_i'\beta/\sigma))+1(y_i > 0)(\log\phi((y_i-x'\beta)/\sigma)-\log\sigma).$$

- The computation is easy because the log likelihood turns out to be concave.
- The Stata command is `tobit y x1 x2 ... xK, ll(0)`. Here '`ll(0)`' indicates that the lower limit is at zero.

Reporting the estimation results

- We obtain estimates $\hat{\beta}_j$ along with standard errors. We also obtain $\hat{\sigma}$ and its standard error.
- We can report estimated PEAs on, say $\mathrm{E}(y \mid x)$. Recall

$$\frac{\partial \mathrm{E}(y \mid x)}{\partial x_j} = \Phi(x'\beta/\sigma)\beta_j = \Pr(y > 0 \mid x)\beta_j.$$

  Hence

$$\Phi(\bar{x}'\hat{\beta}/\hat{\sigma})\hat{\beta}_j$$

  for continuous variables.
- The estimated APEs are

$$\left(n^{-1}\sum_{i=1}^{n}\Phi(x_i'\hat{\beta}/\hat{\sigma})\right)\hat{\beta}_j.$$

- For the standard error of PEA or APE, use bootstrapping or delta method.

- For discrete explanatory variables or for large changes in continuous ones, we can compute the difference in $\mathrm{E}(y \mid x)$ at different values of x.
- For example, let x_K be a binary variable. Define $\hat{w}_{i1} = x_{i(K)})'\hat{\beta}_{(K)} + \hat{\beta}_K$ and $\hat{w}_{i0} = x'_{i(K)}\hat{\beta}_{(K)}$, where $x_{i(K)} = (x_{i1}, \ldots, x_{i,K-1})'$.
- Then the average difference

$$n^{-1} \sum_{i=1}^{n} ([\Phi(\hat{w}_{i1}/\hat{\sigma})\hat{w}_{i1} + \hat{\sigma}\phi(\hat{w}_{i1}/\hat{\sigma})] - [\Phi(\hat{w}_{i0}/\hat{\sigma})\hat{w}_{i0} + \hat{\sigma}\phi(\hat{w}_{i0}/\hat{\sigma})])$$

is the partial effect on $\mathrm{E}(y \mid x)$, though it is usually called *average treatment effect* (ATE).

Goodness of fit

- Recall the formula of $\mathrm{E}(y \mid x)$,

$$\mathrm{E}(y \mid x) = \Phi(x'\beta/\sigma)x'\beta + \sigma\phi(x'\beta/\sigma).$$

- It is reasonable to view

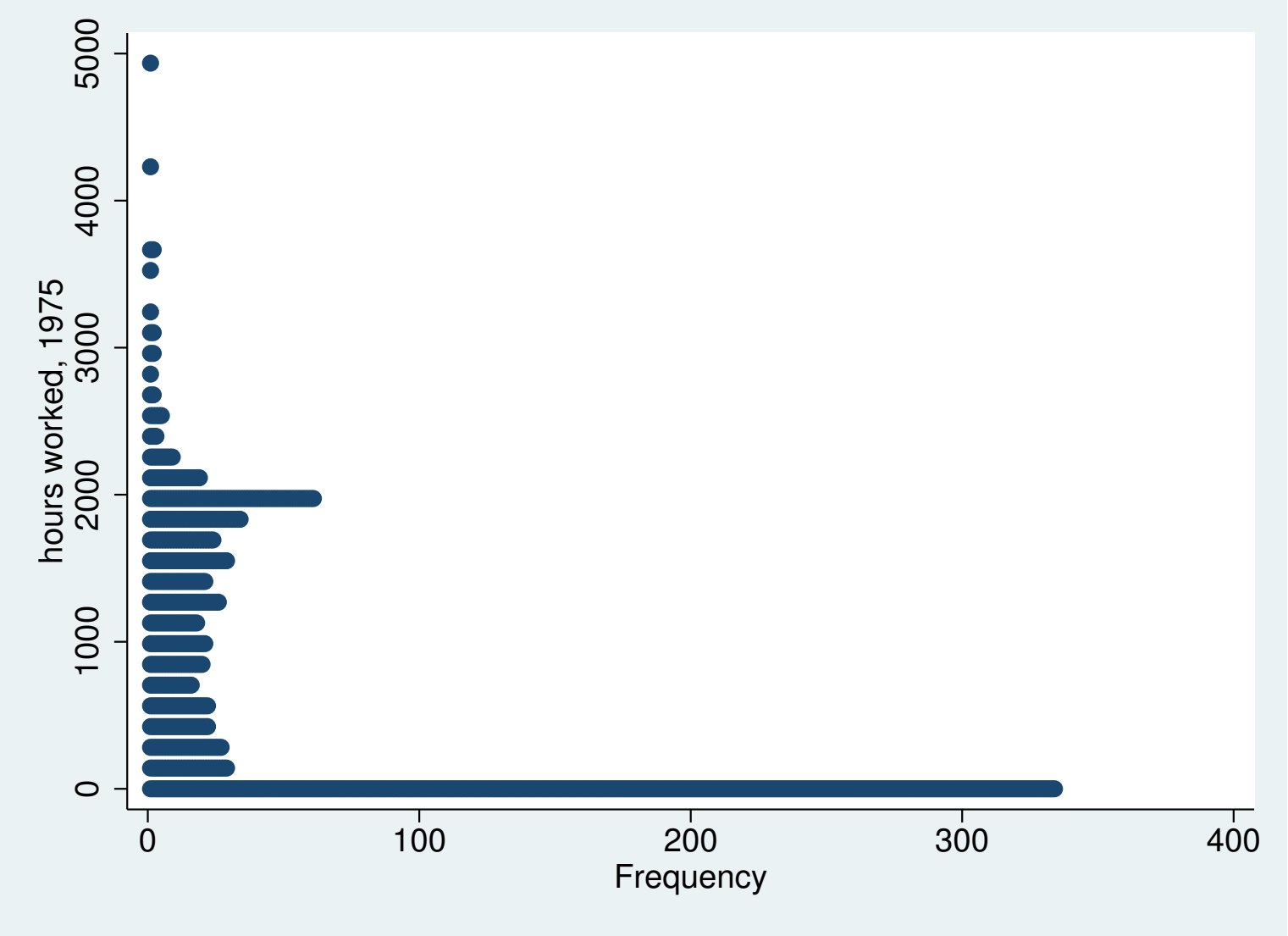
$$\hat{y}_i \equiv \hat{\mathrm{E}}(y_i|x_i) = \Phi(x_i'\hat{\beta}/\hat{\sigma})x_i\hat{\beta} + \hat{\sigma}\phi(x_i'\hat{\beta}/\hat{\sigma}),$$

  as an estimate of $\mathrm{E}(Y \mid X = x_i)$.
- The squared correlation between y_i and $\hat{y}_i$ is a reasonable goodness of fit measure.
- Of course, given $\hat{y}_i$, we can define sum of squared residuals-type R squared.

Female labor supply example

- The data is from Mroz (1987, *Econometrica*).
- y is "annual hours of working"
- x includes "spouse income", "years of schooling", "working experience (exper)", "exper squared", "age", "number of kids $<$ 6 years", "number of kids 6—18" and constant.

Annual working hours in sample



```
. sum hours

    Variable |       Obs        Mean    Std. Dev.       Min        Max
-------------+--------------------------------------------------------
       hours |       753    740.5764    871.3142          0       4950


. count if hours == 0
  325
```

```
. tobit hours nwifeinc educ exper expersq age kidslt6 kidsge6, ll(0)

Tobit regression                                  Number of obs   =        753
                                                  LR chi2(7)      =     271.59
                                                  Prob > chi2     =     0.0000
Log likelihood = -3819.0946                       Pseudo R2       =     0.0343

------------------------------------------------------------------------------
       hours |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
    nwifeinc |  -8.814243   4.459096    -1.98   0.048    -17.56811   -.0603724
        educ |   80.64561   21.58322     3.74   0.000     38.27453    123.0167
       exper |   131.5643   17.27938     7.61   0.000     97.64231    165.4863
     expersq |  -1.864158   .5376615    -3.47   0.001    -2.919667   -.8086479
         age |  -54.40501   7.418496    -7.33   0.000    -68.96862    -39.8414
     kidslt6 |  -894.0217   111.8779    -7.99   0.000    -1113.655   -674.3887
     kidsge6 |    -16.218   38.64136    -0.42   0.675    -92.07675    59.64075
       _cons |   965.3053   446.4358     2.16   0.031     88.88528    1841.725
-------------+----------------------------------------------------------------
      /sigma |   1122.022   41.57903                      1040.396    1203.647
------------------------------------------------------------------------------
  Obs. summary:        325  left-censored observations at hours<=0
                       428     uncensored observations
                         0  right-censored observations
```

- To compute the goodness of fit, we need $\hat{y}_i$,

$$\hat{y}_i \equiv \hat{\mathrm{E}}(y_i|x_i) = \Phi(x_i'\hat{\beta}/\hat{\sigma})x_i\hat{\beta} + \hat{\sigma}\phi(x_i'\hat{\beta}/\hat{\sigma}).$$

- `predict` *`newvar`*`, xb` creates `newvar` containing $x'\hat{\beta}$.
- We also need $\hat{\sigma}$, which is stored as variable `_b[/sigma]`.
- See also UCLA Stata FAQ for how to access information stored after running Stata commands.

```
. predict xbh, xb

. gen hoursh = normal(xbh/_b[/sigma])*xbh + _b[/sigma]*normalden(xbh/_b[/sigma])

. sum hours hoursh

    Variable |       Obs        Mean    Std. Dev.       Min        Max
-------------+--------------------------------------------------------
       hours |       753    740.5764    871.3142          0       4950
      hoursh |       753    721.4201    473.6053   3.496456   1993.885

. corr hours hoursh
(obs=753)

             |    hours   hoursh
-------------+------------------
       hours |   1.0000
      hoursh |   0.5237   1.0000

. di .5237^2
.27426169

. * This squared correlation can be used as an R-squared to compare with other
. * models, including a linear model.
```

```
. reg hours hoursh, nocons

      Source |       SS       df       MS              Number of obs =     753
-------------+------------------------------           F(  1,   752) = 1031.68
       Model |   569084372     1   569084372           Prob > F      =  0.0000
    Residual |   414810722   752  551610.002           R-squared     =  0.5784
-------------+------------------------------           Adj R-squared =  0.5778
       Total |   983895094   753  1306633.59           Root MSE      =   742.7

------------------------------------------------------------------------------
       hours |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
      hoursh |   1.007564    .031369    32.12   0.000     .9459831    1.069146
------------------------------------------------------------------------------

. * Note how close the coefficient is to one; this at least suggests the Tobit
. * provides sensible estimates of the "unconditional" mean.
```

Partial effect needs to be scaled in Tobit

$$\frac{\partial \mathrm{E}(y \mid x)}{\partial x_j} = \Phi(x'\beta/\sigma)\beta_j.$$

APE estimates:

$$n^{-1} \sum_{i=1}^{n} \Phi(x_i'\hat{\beta}/\hat{\sigma})\hat{\beta}_j.$$

The following are APE of education and the number of kids below 6 years old.

```
. qui tobit hours nwifeinc educ exper expersq age kidslt6 kidsge6, ll(0)

. gen scale = normal(xbh/_b[/sigma])

. sum scale

    Variable |       Obs        Mean    Std. Dev.       Min        Max
-------------+--------------------------------------------------------
       scale |       753    .5886634    .2426614   .0092704    .960908

. di .589*80.65
47.50285

. di .589*(-894.02)
-526.57778
```

Bootstrap to calculate the standard error of APE when ‘kidslt6’ jumps from zero to 1 and to 2. The program is from Wooldridge.

```
. do ex17_2_boot
. capture program drop tobit_boot
. program tobit_boot, rclass
  1.
. tobit hours nwifeinc educ exper expersq age kidslt6 kidsge6, ll(0)
  2.
. predict xbh, xb
  3. gen scale = normal(xbh/_b[/sigma])
  4. gen pe1=scale*_b[educ]
  5. sum pe1
  6. return scalar ape1=r(mean)
  7. gen xbh0 = xbh - _b[kidslt6]*kidslt6
  8. gen xbh1 = xbh0 + _b[kidslt6]
  9. gen xbh2 = xbh0 + _b[kidslt6]*2
 10. gen m2 = normal(xbh2/_b[/sigma])*xbh2 + _b[/sigma]*normalden(xbh2/_b[/sigma])
 11. gen m1 = normal(xbh1/_b[/sigma])*xbh1 + _b[/sigma]*normalden(xbh1/_b[/sigma])
 12. gen m0 = normal(xbh0/_b[/sigma])*xbh0 + _b[/sigma]*normalden(xbh1/_b[/sigma])
 13. gen pe2 = m1 - m0
 14. sum pe2
 15. return scalar ape2=r(mean)
 16. gen pe3 = m2 - m1
 17. sum pe3
 18. return scalar ape3=r(mean)
 19.
. drop xbh scale xbh1 xbh0 xbh2 m2 m1 m0 pe1 pe2 pe3
 20. end
```

```
.
. bootstrap r(ape1) r(ape2) r(ape3), reps(500) seed(123): tobit_boot
(running tobit_boot on estimation sample)

Bootstrap results                               Number of obs      =       753
                                                Replications       =       500

      command:  tobit_boot
        _bs_1:  r(ape1)
        _bs_2:  r(ape2)
        _bs_3:  r(ape3)

------------------------------------------------------------------------------
             |   Observed   Bootstrap                         Normal-based
             |      Coef.   Std. Err.      z    P>|z|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
       _bs_1 |   47.47311   13.04099     3.64   0.000     21.91324    73.03299
       _bs_2 |   -487.213   54.03773    -9.02   0.000     -593.125    -381.301
       _bs_3 |  -246.1717   12.31618   -19.99   0.000    -270.3109   -222.0324
------------------------------------------------------------------------------

.
. program drop tobit_boot

.
end of do-file
```

```
. reg hours nwifeinc educ exper expersq age kidslt6 kidsge6, robust

Linear regression                                      Number of obs =     753
                                                       F(  7,   745) =   45.81
                                                       Prob > F      =  0.0000
                                                       R-squared     =  0.2656
                                                       Root MSE      =  750.18

------------------------------------------------------------------------------
             |               Robust
       hours |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
    nwifeinc |  -3.446636   2.240662    -1.54   0.124    -7.845398    .9521268
        educ |   28.76112   13.03905     2.21   0.028     3.163468    54.35878
       exper |   65.67251   10.79419     6.08   0.000     44.48186    86.86316
     expersq |  -.7004939   .3720129    -1.88   0.060    -1.430812    .0298245
         age |  -30.51163   4.244791    -7.19   0.000    -38.84481   -22.17846
     kidslt6 |  -442.0899   57.46384    -7.69   0.000    -554.9002   -329.2796
     kidsge6 |  -32.77923   22.80238    -1.44   0.151     -77.5438    11.98535
       _cons |   1330.482   274.8776     4.84   0.000     790.8556    1870.109
------------------------------------------------------------------------------

. * Linear regression coefficient on kidslt6 is very close to the weighted
. * average of the Tobit effects.
. * But the linear regression coefficient for educ (28.8) is well below the APE
. * from the Tobit model (47.5).
```

Specification issues

Neglected heterogeneity

- Suppose the true model is

$$y = \max(0, x'\beta + \gamma q + u), \quad u|x, q \sim \mathcal{N}(0, \sigma^2), \quad q \perp\!\!\!\perp x, \quad q \sim \mathcal{N}(0, \tau^2).$$

- Suppose we somehow ignored q. The model we estimate becomes

$$y = \max(0, x'\beta + v), \quad v = \gamma q + u, \quad v|x \sim \mathcal{N}(0, \sigma^2 + \gamma^2\tau^2).$$

  Clearly, this is still a Tobit model, but the underlying variance is different.

- Ignoring q and running Tobit will consistently estimate β and $\eta^2 \equiv \sigma^2 + \gamma^2\tau^2$.

- How about partial effect? I really don't care about β or σ.
- Consider the following APE,

$$\mathrm{E}_q\left[\frac{\partial \mathrm{E}(y \mid x_0, q)}{\partial x_j}\right].$$

  Partial effects of x_j on $\mathrm{E}(y \mid x)$ evaluated at x_0. Average across q.
- Since $\mathrm{E}(y \mid x_0) = \mathrm{E}(\mathrm{E}(y \mid x_0, q) \mid x_0) = \mathrm{E}_q[\mathrm{E}(y \mid x_0, q)]$, we have

$$\frac{\partial \mathrm{E}(y \mid x_0)}{\partial x_j} = \mathrm{E}_q\left(\frac{\partial \mathrm{E}(y \mid x_0, q)}{\partial x_j}\right).$$

  Thus, we estimate the desired APEs by simply *ignoring* the heterogeneity.

Endogenous explanatory variables

- Consider the following model with a continuous endogenous explanatory variable:

$$y_1 = \max(0, z_1'\delta_1 + \alpha_1 y_2 + u_1)$$
$$y_2 = z'\delta_2 + v_2 = z_1'\delta_{21} + z_2'\delta_{22} + v_2.$$

  Here (u_1, v_2) are zero-mean normally distributed, independent of z. If u_1 and v_2 are correlated, then y_2 is endogenous.

- Besides δ_1 and α_1, we are interested in estimating APEs. For example,

$$\mathrm{E}_{u_1}\left(\max(0, \bar{z}_1'\delta_1 + \alpha_1(\bar{y}_2 + 1) + u_1) - \max(0, \bar{z}_1'\delta_1 + \alpha_1\bar{y}_2 + u_1)\right)$$

- Consider the linear projection of u_1 onto v_2,

$$u_1 = \theta_1 v_2 + e_1,$$

where $\theta_1 = \eta_1/\tau_2^2$, $\eta_1 = \text{Cov}(u_1, v_2)$, $\tau_2^2 = \text{Var}(v_2)$, and e_1 is independent of v_2 and follows mean-zero normal distribution.

- We now have

$$y_1 = \max(0, z_1'\delta_1 + \alpha_1 y_2 + \theta_1 v_2 + e_1), \quad e_1|z, v_2 \sim \mathcal{N}(0, \tau_1^2).$$

Provided that v_2 is known/estimable, the above display is a Tobit model.

- We have the following two step procedure by Smith and Blundell (1986):
  1. Estimate the reduced form of y_2 by OLS and save the residuals $\hat{v}_2$.
  2. Estimate the standard Tobit model of y_1 on $z, y_2, \hat{v}_2$. We obtain consistent estimators of $\delta_1, \alpha_1, \theta_1, \tau_1^2$.
- Endogeneity can be tested by $H_0 : \theta_1 = 0$.
- This is a two-step procedure. Hence we need to adjust the standard error of the second step taking account of the estimation error of $\hat{v}_2$. Bootstrap can also be applied to calculate the standard error.